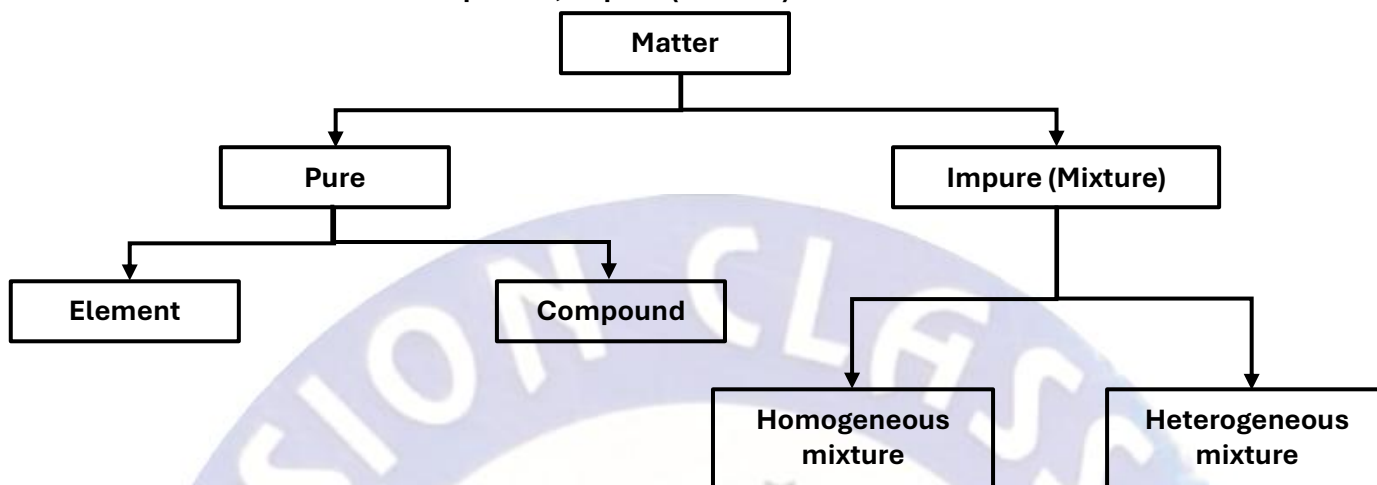


EXPLORATION OF MIXTURE AND THEIR SEPARATION

- **Introduction**

- In this chapter, we study about classification of matter on the basis of chemical properties.
- All the matter around us is either pure or, impure (mixture).



- **Pure Substance**

- A pure substance is one which is made up of only one kind of particles.

For example:-

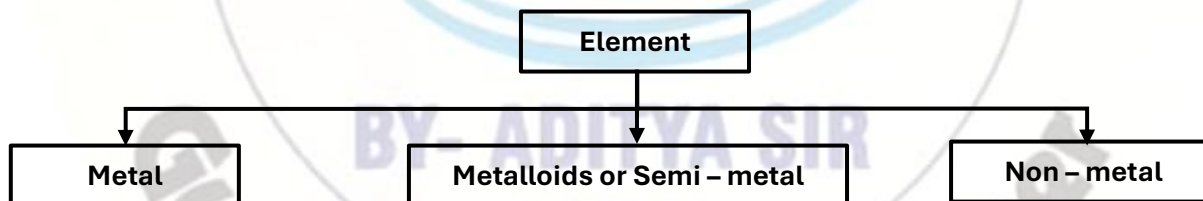
- ✓ Elements and compounds which are made by same kind of particle atoms or, molecules.

- **Element**

- An element is the purest form of matter which is made up same kinds of atoms.
- An element which can't be broken into two or, more parts by a physical or simple chemical process.

For example:-

- ✓ Hydrogen, Carbon, Oxygen, Sodium, Magnesium, Calcium, Gold, Silver, Iron etc.



Note:

- There are 118 elements exist in which atomic no. 1 to 94 exist in nature and 24 elements are prepared in laboratory.

- **Metals**

- Those elements which have 1 to 3 electrons in their outermost shell, are called metals.

Exception:-

- ✓ Hydrogen & Helium are non – metals.

- Metals having generally 'um' suffix.

For example:-

- ✓ Sodium, Magnesium, Aluminum, Potassium, Calcium, Gold (Aurum), Silver, Mercury etc.

▪ **Properties of metals**

Case 1: Physical properties of metal

a) These are good conductor of electricity and heat because they have free electrons.

Note:

- Silver is the best conductor of electricity and lead is least conductor of electricity.

b) Metals are solid at room temperature (at 25°C).

Exception:

- ✓ Mercury (Hg) in liquid.

c) Metals are hard and rigid.

Exception:

- ✓ Sodium & Potassium are soft which are cut by knife.

d) The density of metal is high because, mass (↑) & volume (↓).

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

e) The boiling and melting points of metals are high because their atoms are bonded strongly.

f) Metals are shiny which is known as metallic lustre. So, metals are lustrous.

g) Metals are ductile i.e, they can be made in wire by stretching.

For example:-

- ✓ 2 km long wire can be formed by 1g gold.

h) Metals are malleable i.e, metal can be made into sheet, axe etc by hammering.

i) Metals are usually strong thus, they have high tensile strength.

j) Metals are sonorous.

Case 2: Chemical Properties of metal will be study in class – 10th

• **Non – Metal**

- That element which has 5 to 8 electrons in their outer most shell are called non – metal.

Exception:-

- ✓ Carbon has 4e⁻(s) in outer shell.

- Total no. of Non – metals is 18.

For example:-

- ✓ Hydrogen → H (g)
✓ Carbon → C (s)
✓ Nitrogen → N (g)
✓ Oxygen → O (g)
✓ Fluorine → F (g)
✓ Phosphorus → P (s)
✓ Sulphur → S (s)
✓ Bromine → Br (l)
✓ Iodine → I (s)
✓ Selenium → Se (s)

All inert gases

Helium → He

Neon → Ne

Argon → Ar

Krypton → Kr

Xenon → Xe

Radon → Rn

▪ **Properties of Non – metals**

Case 1: Physical properties of non – metal

i) Non – metal is non – Conductor or, insulator of electricity.

exception:

- ✓ Graphite is conductor of electricity.

ii) Non – metal are solid or, liquid or gas at room temperature.

iii) Non – metal have less hardness than metal.

exception:

✓ Diamond is a non – metal which is hardest substance in the universe.

iv) Non – metals have comparatively low melting points and boiling points.

v) The density of Non – metal is lower than metal.

vi) Non – metals are not lustrous.

exception:

✓ Graphite and iodine.

vii) Non – metals are brittle.

viii) Non – metals are not ductile.

ix) Non – metals are not sonorous.

x) Non – metals have low tensile strength.

Case 2: Chemical properties of Non – metal will be studied in class – 10th

▪ Difference between metal and Non – metal

Case 1: On the basis of physical properties

Metals	Non – Metals
Metals are solid at room temperature.	Non – metals exist in all three states of matter.
Metals are lustrous.	These are not lustrous.
Metals are good conductors of heat and electricity.	Non – metals are poor conductors of electricity.
The density of metal is quite high.	The density of non – metals is low.
Metals are sonorous.	Non – metals are not sonorous.
Metals are malleable and ductile.	Non – metals are brittle.
Metals have normally high M.P & B.P.	The M.P and B.P of non – metals are usually low.

• Metalloids OR Semi – metals

➤ Those elements whose properties lies between metals and Non – metals are called metalloids.

➤ There are 8 metalloids

i) Boron

ii) Silicon

iii) Germanium

iv) Arsenic

v) Antimony

vi) Tellurium

vii) Polonium

viii) Astatine

• Compound

➤ Compound is the purest substance of matter which are made by two or, more elements (unlike).

For example:-

✓ Hcl, Nacl, Na₂SO₄, HNO₃, CO₂, H₂O etc.

▪ Properties of a compound

a) The elements of a compound are in the fixed ratio by mass/no. of molecule/weight.

b) Compound can exist in free state.

c) Compound may or, may not conductor of electricity.

d) The smallest particle of compound is molecule.

e) It can be broken into two or more parts.

f) The properties of a compound are entirely different from its constituents.

g) A compound has a fixed M.P & B.P.

- Only for understand

$\therefore$ The ratio of Hydrogen and Oxygen is 1:8 by weight.

∴ The ratio of Carbon and Oxygen is 3:8 by weight.

3. Compound $\left\{ \begin{array}{l} \text{H}_2\text{O} \\ 2 \text{H}_2\text{O} \\ 3 \text{H}_2\text{O} \\ 4 \text{H}_2\text{O} \\ 100 \text{H}_2\text{O} \end{array} \right. \rightarrow \begin{array}{l} \text{1 mole of water} \\ \text{2 mole of water} \\ \text{3 mole of water} \\ \text{4 mole of water} \\ \text{100 mole of water} \end{array}$

Compound

1. **100 H₂O** → Called Compound
→ Not called molecule
2. **Cl₂, O₂, O₃, P₄, S₈** etc. are molecules not compound.
3. **C₁₂H₂₂O₁₁** is a molecule of compound sugar.

- **Differences between element and compounds**

Element	Compound
It can't be made by any physical and chemical process.	It is made by chemical combination of two or more element in the fixed ratio by weight.
It is made by same type of atoms.	It is made by two or more different elements.
It can't be split into two or more substances.	It can be split into two or more substances.
It is conductor of electricity.	It may or may not be conductor of electricity.
The smallest particle of element is called an atom.	The smallest particle of compound is called molecules.

- **Mixture (Impure substance)**

- Those substances which are made by combination of two or more elements/compounds, are called mixture.
- So, the components/Constituents of a mixture can be separated by any simple physical process.

For example:-

- ✓ Air is a mixture of nitrogen, oxygen, carbon dioxide, water vapor (moisture) etc.
- ✓ Gunpowder is a mixture of potassium nitrate, Sulphur and coal.
- ✓ Steel is a mixture of iron, carbon, nickel, chromium, magnesium and vanadium.

Types of Mixture

1. Homogeneous mixture.
2. Heterogeneous mixture.

1. Homogeneous mixture

- Those mixtures which are made by the combination of substances in a certain ratio, are called homogeneous mixtures.
- All types of homogeneous mixtures are called solutions.
- In Homogeneous mixtures, the components are completely mixed together such that indistinguishable from one – another.

For example:-

- ✓ The water and salt mixture (solution), the sugar and the water mixture (solution), The copper sulphate (CuSO_4) solution Petrol and oil mixture, Soft drink, Soda water, Air, Brass, Kerosene oil, petrol, beer, Liquor etc.

2. Heterogeneous mixture

- Those mixtures which are made by two or more substances in any properties i.e, non – uniform composition of constituents.
- It has visible boundaries of separation between the constituents.

For example:-

- ✓ Gunpowder, Fog, Sand and water mixture, Sand and Salt mixture, oil and water mixture.

Note:

- Colloids and suspension are also heterogeneous mixture like as, polluted air, Milk, ink, paint, petroleum, wood, Soil, Glass Coal, blood, Milk of magnesia, butter, cheese, Face cream, Shaving cream, Hair Spray etc.

▪ Properties of a mixture

- a) A mixture can be separated into its constituents by physical process like as – Filtration, evaporation, sublimation, etc.
- b) A mixture shows the properties of all the constituents present in it.
- c) Energy (Either in form of heat or light) is usually neither given out nor absorbed in the preparation of a mixture.
- d) The composition of a mixture is variable, the constituents can be presented in any proportion by mass.
- e) A mixture does not have a definite melting point, boiling point etc.
- f) A mixture is usually heterogeneous exception – solution and alloy.

▪ Differences between the compound and mixture

Compound	Mixtures
In it, two or more elements are combined chemically.	In it, two or more elements or compounds are simply mixed & not combined chemically.
A Compound is a pure substance.	A mixture is not a pure substance.
In it, the elements are present in the fixed ratio by mass. This ratio cannot change.	In it, the constituents are not present in the fixed ratio. It can vary.
Compounds are always homogeneous, i.e., they have the same composition throughout.	Mixtures may be either homogeneous or heterogeneous in the nature.
In it, the constituents lose their identities i.e., a compound does not show the characteristics of the constituting elements.	In it, the constituents do not lose their identities, i.e., a mixture shows the characteristics of all the constituents.

In the formation of a compound, energy in the form of heat, light or electricity is either absorbed or evolved.	No energy change is noticed in the formation of a mixture.
In it, the constituent cannot be separated by physical means.	The constituents from a mixture can be separated by physical means in most of the cases.
A compound has fixed melting and boiling point.	The melting and boiling points of a mixture are not fixed.

Q. How will you justify that air is a mixture and not a compound?

OR

Explain why air is considered a mixture and not a compound.

Ans The justification can be done on the basis of the following points in its support.

- Air can be separated into its constituents like oxygen, nitrogen, etc. by the process of fractional distillation.
 - Air shows the properties of all the gases present in it.
For example:-
 - ✓ Oxygen supports combination and air also supports combustion; carbon dioxide turns lime water milky.
 - Heat and light are neither given out nor absorbed when air is prepared by its constituents.
 - Air has a variable composition because air at different places contains different amounts of various gases.
 - Liquid air does not have a fixed boiling point.
- According to the above result, Air is mixture not a compound.

Q. Explain why, water is a compound and not a mixture.

Ans Water is considered as a compound because of following points.

- Water can not be separated into its constituents.
- The properties of water are entirely different from those of its constituents.
For example:-
 - ✓ Water does not burn while Hydrogen burns with support oxygen.
- Heat and light are given out when water is prepared by burning hydrogen in oxygen.
- The composition of water is fixed it contains Hydrogen and oxygen combined together in a fixed proportion of 1:8 by mass. It has a definite formula, H_2O .
- Water has a fixed boiling point of $100^\circ C$ under the S.T.P.

Note:

➤ S.T.P → Standard temperature and pressure.

➔ According to the above facts it is clear that water is a compound not a mixture.

▪ Types of the mixture

1. Solid – Solid mixture

➤ That mixture which components are solid substances are called Solid – Solid mixture.

For example:-

- ✓ A mixture of rice and stone.
- ✓ A mixture of Sodium chloride and ammonium chloride.
- ✓ A mixture of Sand and Sugar.
- ✓ A mixture of Salt and Sugar.

2. Solid – Liquid mixture

- That mixture which components are solid as well as liquid.

For example:-

- ✓ A mixture of sugar, salt and water.
- ✓ A mixture of alcohol and ice.
- ✓ A mixture of wine and ice.

3. Solid – gas mixture

- That mixture which components are solid as well as gas.

For example:-

- ✓ Air is intrapped in the pores of soil particles.

4. Liquid – gas mixture

- That mixture which components are liquid as well as gas.

Note:

- All gases are Partially miscible with liquid. ★

For example:-

- ✓ Cold drink (Carbonic acid).

5. Gas – Gas mixture

- That mixture which both components are gases.

For example:-

- ✓ All gases are mixed in other gases in any proportional as air.

6. Liquid – Liquid mixture

- That mixture which both components are liquids.

For example:-

- ✓ Kerosene is mixed in diesel.
- ✓ Mixture of alcohol and water, milk and water etc.

- Alloy

- Alloy is mixture of either metal and metal or, metal and other substances (non – metal).

For example:-

- ✓ Brass which is made of copper and zinc.
- ✓ Nichrome is made by nickel, chromium, iron and cobalt.
- ✓ Stainless steel is made by chromium, nickel, molybdenum manganese, Aluminum, Carbon, iron etc.

- Advantages of alloy

- It is lighter than its constituents.
- It is harder than its constituents.
- Decreases the rate of corrosion.
- Alloy looks attractive and pretty.

- Disadvantages of alloy

- It is not malleable. Due to its hardness when we strike a hammer then it brakes.
- The boiling and melting point of alloy is higher than its constituents.

- **Separation of components of the mixture**

➤ There are following processes to separate the components of a mixture.

1. **Hand Picking**

➤ The bigger components of a mixture are visible so it can be separated by hand picking or using forceps.

For example:-

✓ Piece of stone may be separated from rice by hand picking.

2. **Sieving**

➤ A Sieve is a mechanical device in which mesh is attached to a frame, when the mixture is placed on the mesh and shaken, the particles smaller than the holes of mesh pass through but larger one's stay on the mesh.

For example:-

✓ The sieve is used in the kitchen, construction site.

3. **Sedimentation**

➤ If a mixture of liquid and insoluble solid is allowed to stand still for some time, the solid particles are settle down at the bottom. This process is called sedimentation.

➤ The solid which is settled down at the bottom is called sediments.

For example:-

✓ The dirt particles are present in muddy water can be separated by sedimentation process.

✓ The mixture of sand water etc.

✓ The mixture of immiscible liquids.

- **Decantation**

➤ After the dirt particles settle down at the bottom, water is poured into another container. The process of separating a liquid from the sediment by carefully pouring in into another container is known as "decantation".

4. **Centrifugation/Churning**

➤ Centrifugation is a process used when the sedimentation of liquids having suspended particles taken place very slowly. So, mixture is taken into specially designed test tubes known as centrifuge tubes. The centrifuge tubes are placed in separated holders.

The holders are then rotated rapidly with the help of the handle. After sometimes the rotation is stopped and centrifuge tubes are taken out. It seems that the solid particles have settled above the surface. How solids are collected by decantation.

For example:-

✓ The cream is separated from milk because cream is lighter than milk.

5. **Filtration**

➤ If the process of sedimentation fails to separate very minute solid particle from the mixture, in this cases complete separation is done by filtration, the liquid is allowed to pass through a filter paper.

The filtered liquid called filtrate and the solid particles are collected on the filter paper called residue.

For example:-

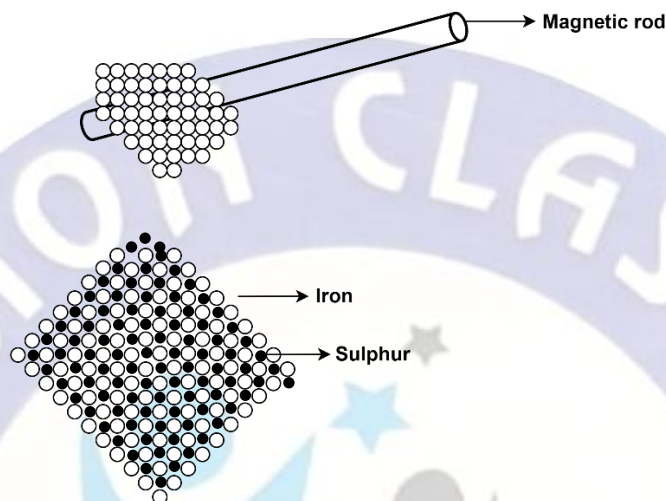
✓ The mixture of sand and water.

6. Magnetic separation

- This process is only applicable for those mixture which one of the components is magnetic in nature. Thus, we separated the magnetic substance by the heavy for magnet.

For example:-

- ✓ We know that in ferrous sulphate (FeSO_4), the iron particle is mixed in the Sulphur. How a bar magnet is moved above the powder of Ferrous sulphate due to which magnetic substance attracts towards the magnet and non – magnetic part (Sulphur) remains same.

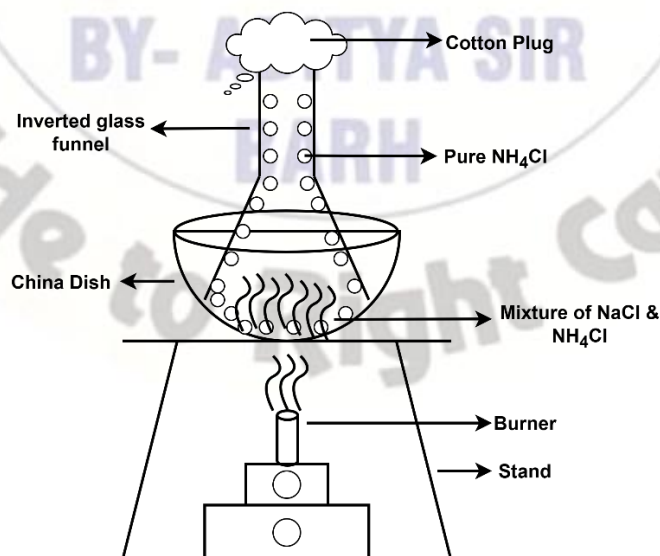


7. Sublimation

- This process is only applicable for those mixtures which one of the component vaporised without melting and on cooling the vaporised components directly change into solids.

For example:-

- ✓ Ammonium chloride can be separated from common salt by this process. In this process we take a mixture of ammonium chloride and common salt in a China dish. The dish is covered with narrow opening of the inverted glass funnel plugged with cotton on heating the China dish ammonium chloride sublimates and collect it in the cotton.



8. Evaporation

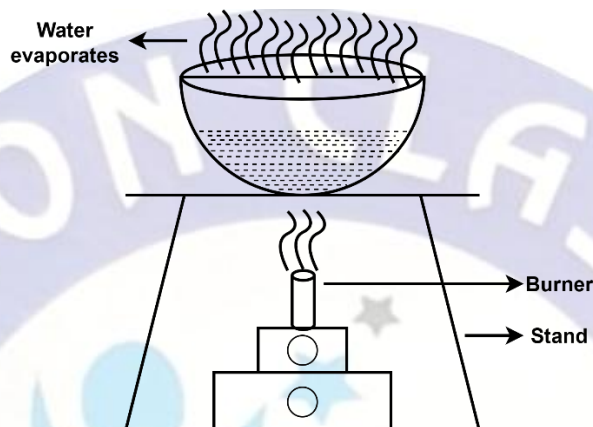
- This process is used for separating a solid (dissolved) from a liquid. First the solution is taken a glass of basin and heated slowly on a water bath. The solvent evaporates away and the solid is left behind the residue.

- **Water bath**

- It is an instrument which contain hot water.

OR, Second method

- The solution of common salt and water is taken in a china dish and heated gently by using a burner. The water present in salt solution will form water vapors and escape into the atmosphere. When all the water present in the solution of common salt and water gets evaporated, then common salt is left behind in the china dish as well as a white solid.



9. **Crystallisation [CRYSTALLISATION]**

- The crystallisation process is used for separating the impurities in the process of evaporation.

In this process first the solution is taken in a china dish and evaporates on the water bath till the solution becomes saturated at the point of saturation the solid particles in a regular or, definite shape begin to appear. These particles are called Crystal.

The saturated solution is removed from china dish and allowed to cool the room temperature (at 25°C). How the crystal of solids separated from the solution. This process is called crystallisation.

For example:-

- ✓ The salt is obtained from sea water contains many impurities. These impurities can be removed by crystallisation process.

Note:

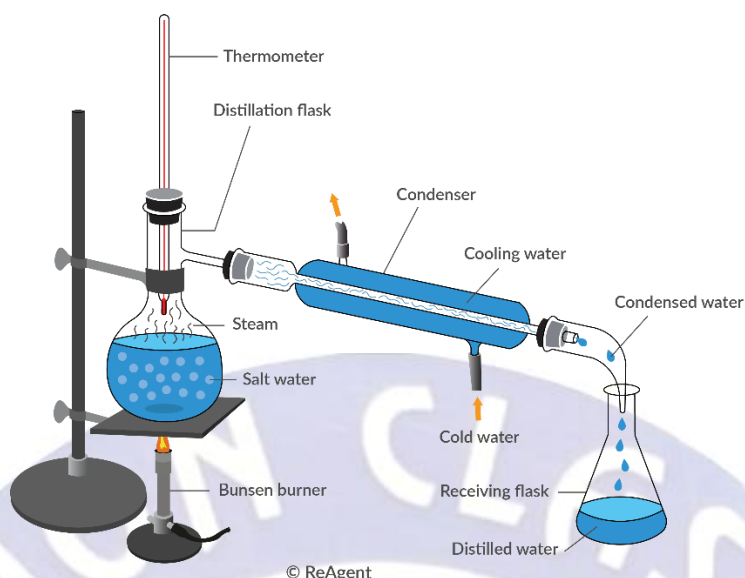
- Sugar crystal can not be obtained by simple vapourisation process.

10. **Distillation**

- This process is used to separating the two miscible liquids which boiling points differ at – least by 25°C.

In the process of distillation, we used a specially designed round – bottom flask is called distillation flask. Thus, the side arm at the neck of the flask is attached to cooling device called lie – big condenser. The end of the condenser is connected to a receiver.

The solution is taken in the distillation flask and heated slowly to boiling, the vapour that comes out pass through the condenser where it is cooled into liquid. The liquid is collected in the receiver. The solid/liquid matter is left behind in the distillation flask.



Note:

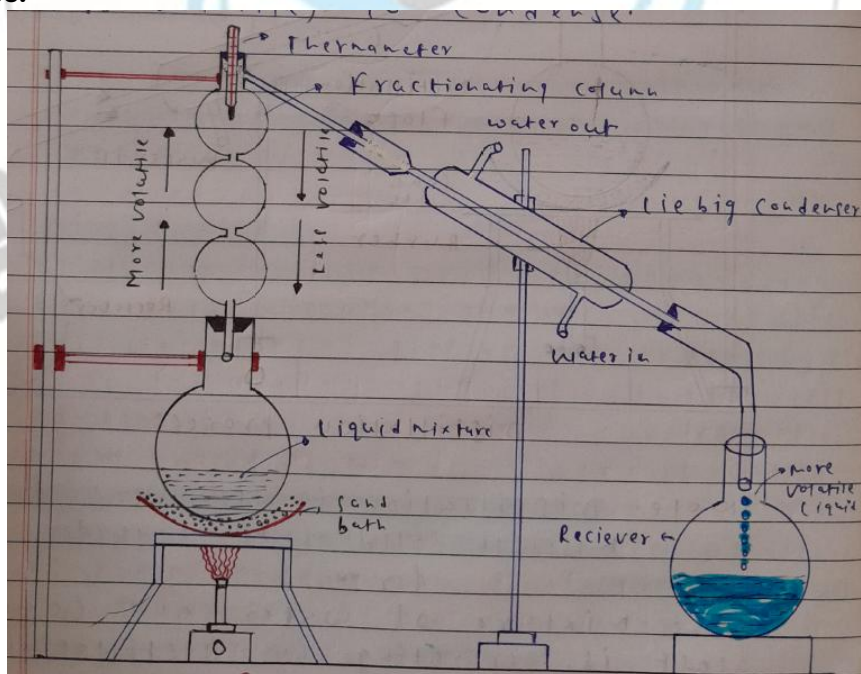
- Distillation is also used for miscible solid into liquid.

For example:-

- ✓ The mixture of water and common salt is separating by distillation process.

11. Fractional Distillation

- This process is used to separate the components of a mixture of two or more miscible liquids whose boiling points differ by 25°C or less than 25°C . In this process the mixture is heated in a round-bottomed flask over a sand bath. The flask is attached to a fractionating column. The fractionating column has several trays to allow the high boiling liquid (less volatile) to condense.



The vapour of lower boiling point liquid (more volatile) goes into the condenser where it is condensed and liquid thus obtained is then collected in the receiver. The temperature remains fixed at the boiling point of the lower boiling liquid when this more volatile liquid is completely distilled, the temperature rises to the boiling point of the next higher boiling liquid. This liquid is collected in the another receiver.

For example:-

- ✓ Methyl alcohol (B.P 65°C) can be separated from acetone (B.P 56°C) by fractional distillation process.

12. Separation by a separating funnel

- A separating funnel is used to separate the components of two immiscible liquids.

For example:-

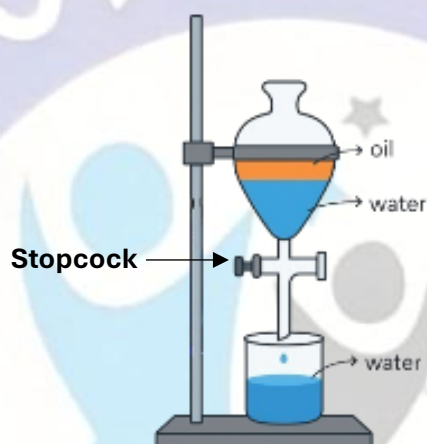
- The mixture of oil and water can be separated by separating funnel.

- **Process of Work:-**

- A separating funnel consists of a glass bulb filled with a stop cock in its stem.

First the mixture is taken in separating funnel and allowed to stand still for some time, the oil being lighter forms the upper layer while water forms the lower layer.

The stopcock is opened and the lower layer of water is carefully drained out in a beaker.



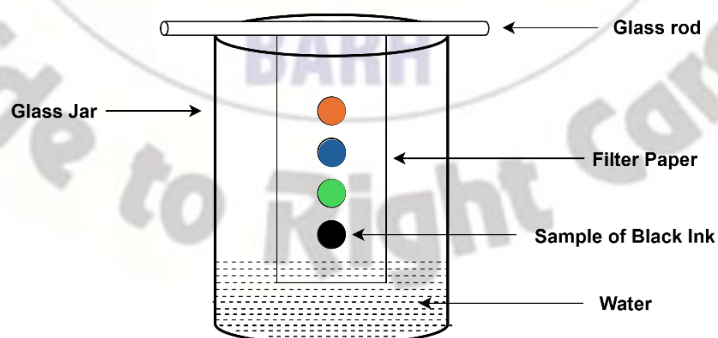
13. Chromatography

- This word is derived from Greek word "**Chroma**".

→ Colour/dye

- Chromatography is that process which is used to separating the dissolving components of dye.

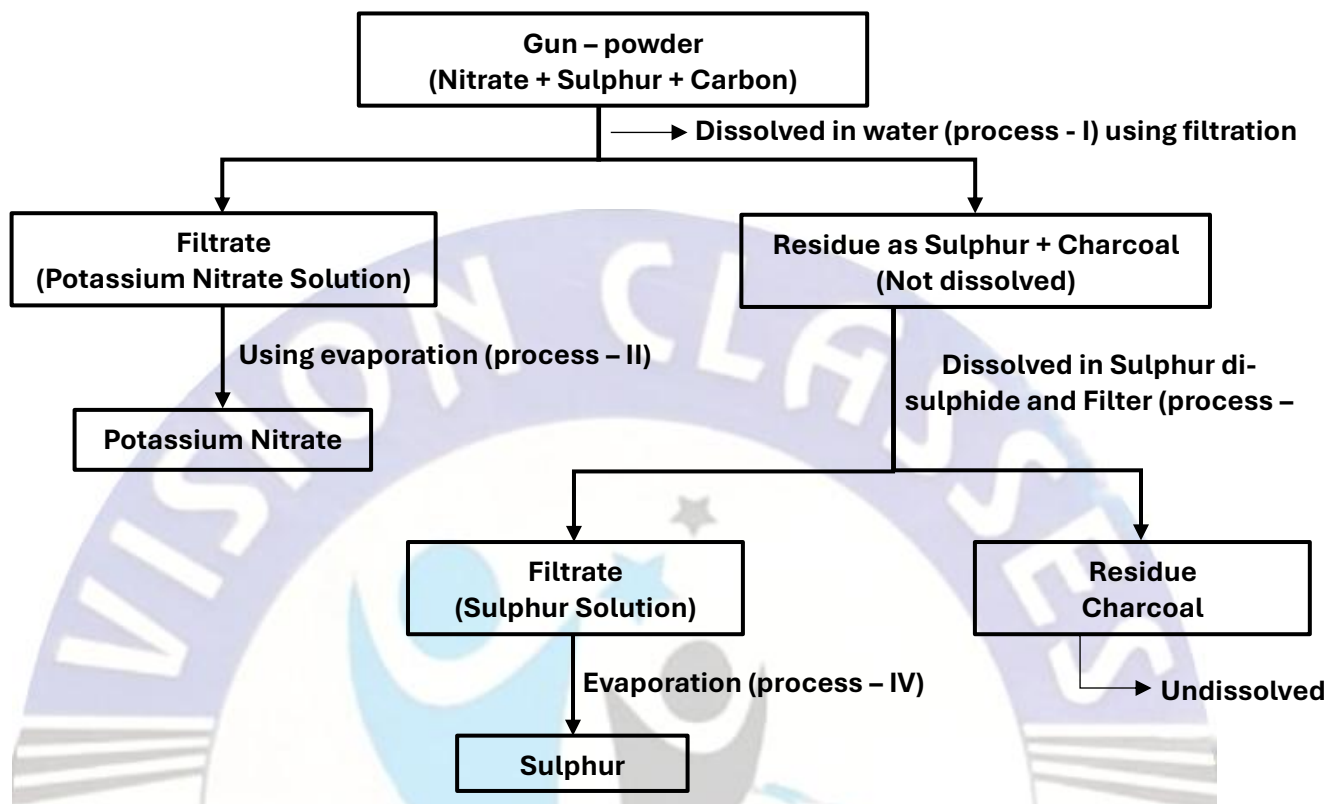
Take a thin strip of filter paper, now place a drop of black ink at a distance of about 2cm from the base of the filter paper. Next dipped the lower end of the filter paper in the water taken in the glass jar. The filter paper should be kept hanging as shown in the figure.



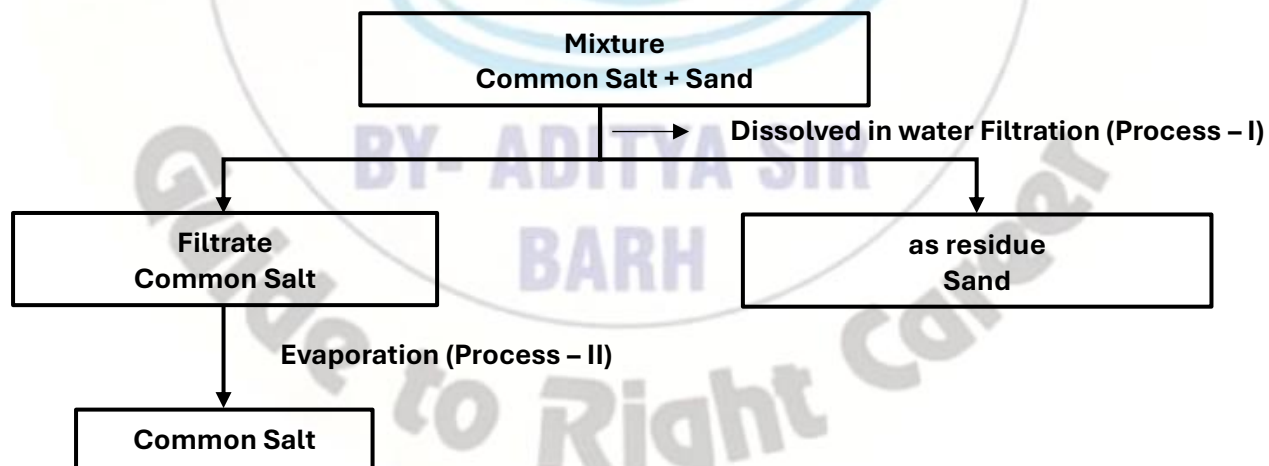
- Now, you will observe spots of different colours rising upto the filter paper. The more soluble component move up faster while the less soluble component follow it.

• **Separate the components of typical mixture**

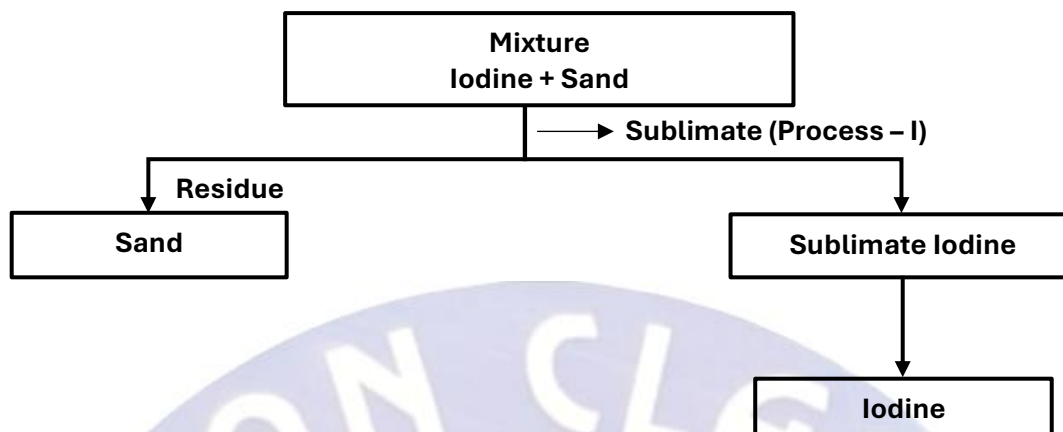
Case 1: To separate the components of Gun – powder



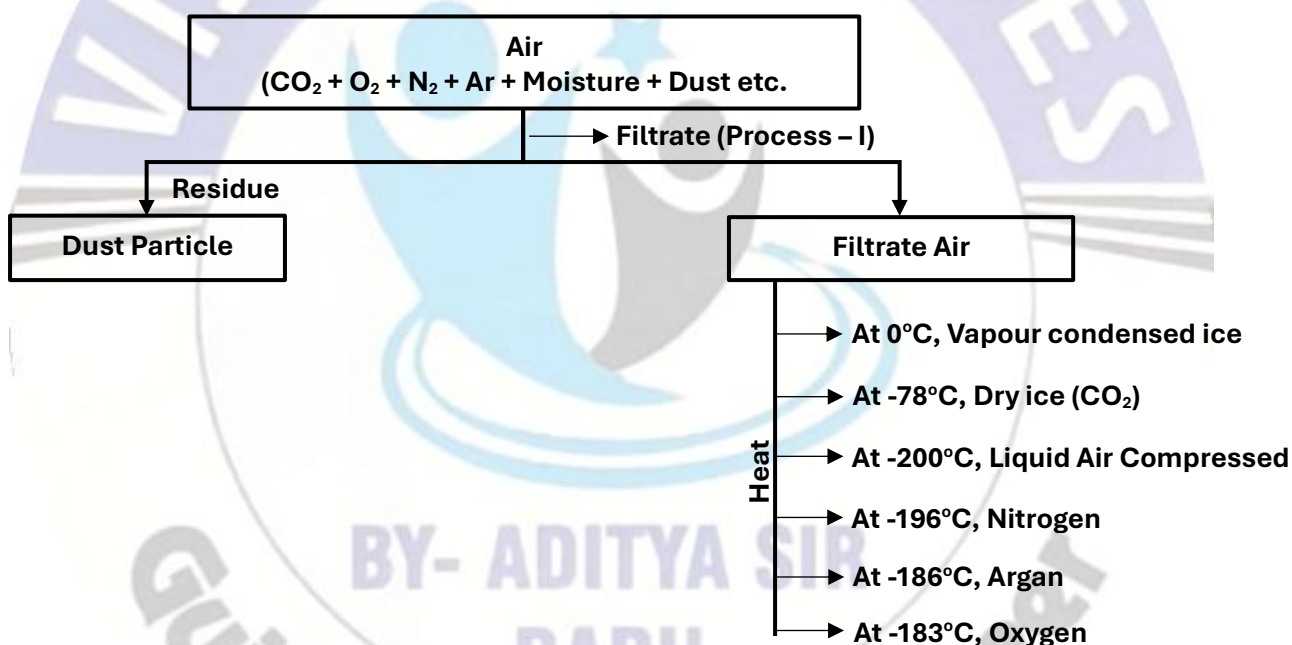
Case 2: To separate the components of Salt and Sand.



Case 3: To separate the component of Sand and Iodine



Case 4: To separate the components of air



• **Purification of water obtained from Pond, Rivers, Lakes etc.**

- The obtained water from ponds, rivers, lakes are not potable.
- To make it potable there are following process used.

Step 1: Sedimentation Tank

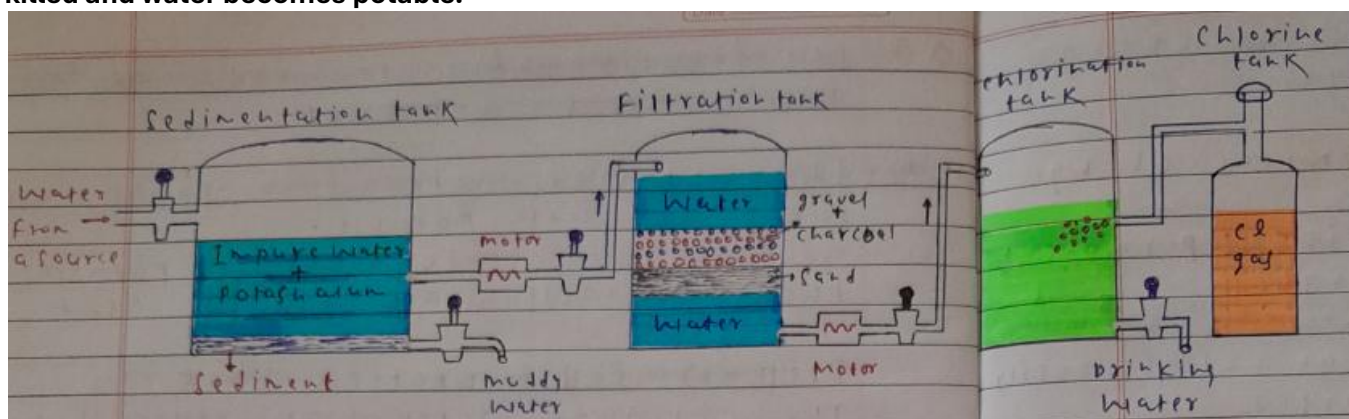
- The obtained water from river, lake etc. is led into sedimentation tank. A small amount of potash alum is added and the water is allowed to stand still due to which muddy substance settled down at the bottom.

Step 2: Filtration Tank

- The clear liquid obtained from sedimentation tank is pumping in filtration tank. Where water has to pass through the layers of charcoal, gravel and fine sand water gets filtered here and becomes free from suspended impurities, this water is led into chlorination tank.

Step 3: Chlorination tank

- In this tank, water is treated with liquid chlorine or, bleaching powder due to which bacterial germs are killed and water becomes potable.



- **Physical Change**

- A physical change is one in which some of the physical properties (colours, shape, state etc.) of the substances undergoing change are temporarily but the actual composition and mass of the substances remains unchanged and no new substances are produced.
- The substance easily stored to its original state as soon at the cause of change is with drawn.

For example:-

- Evaporation of water.
- The dissolution of Common Salt into water.
- The mixture of iron filings and Sulphur.
- The melting ice.
- Folding of paper.
- Melting of wax.
- Stretching of rubber.

Note:

- In Physical Change, no new substances are produced.

- **Chemical Change**

- A chemical change is one in which the substances undergoing change are transformed into new substances whose properties (mass, composition) are entirely different from those of the original substance.
- The new substances can not be changed back to the original substance even if the cause of change is drawback.

Note:

- Chemical change is actually permanent change.

For example:-

- ✓ Burning of coal, souring of milk, Heating of iron and Sulphur together, digestion of food in our body, rusting of iron, mixing of cement and water, ripening of food etc.

- **Difference between Physical Change and Chemical Change**

Physical Change	Chemical Change
It is a temporary change.	It is a permanent change.
It can be easily reversed.	It can not be easily reversed.
Only the physical properties of substance is changed.	Both Physical and Chemical properties are changed.

No new substance is formed in a physical change.	New substance are always formed in a chemical changed.
Identities of the substance do not change.	Identities of the substance changed.
Energy changes normally do not occur.	Energy changes always take place.
Mass of the substance does not change.	Mass of the substance does change.

- **Solution**

- A solution is a homogeneous mixture of two or more substances.

For example:-

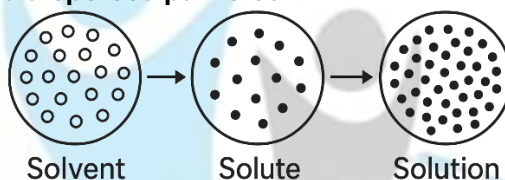
- ✓ Sugar and water solution, Salt and water solution etc.

- **Solvent**

- The liquid in which a substance is dissolved is called a solvent.
- A solvent is a dissolving agent.
- A solvent is also called the medium of dispersion.
- The amount of solvent is generally more than the amount of solute in a solution.

- **Solute**

- The substance which is dissolved in a solvent to produce a solution is called a solute.
- The solute particle is also called dispersed particles.



Note:

- The solute particle hides themselves in the intermolecular space of the solvent.

For example:-

- ✓ The solution of water and sugar (homogeneous).

- **Aqueous Solution**

- The water containing solution is called aqueous solution.
- OR
- The water solvent solution is called aqueous solution.

For example:-

- ✓ The solution of salt and water. It is denoted by NaCl (aq).
- ✓ It is denoted by "aq".

- **Non – aqueous Solution**

- The solution which contain any solvent except water are called non – aqueous solution.

For example:-

- ✓ The solution of alcohol and acetone.

- **True Solution**

- That solution in which the solid particles disappear into the space between the particles of the solvent, called true solution.

For example:-

- ✓ The solution of sugar and water.
- ✓ The solution of salt and water.

Note:

- True solution is simply called solution.

▪ **Properties of Solution/True Solution**

- a) A solution is a homogeneous mixture.
- b) The size of solute particles in a solution is extremely small. It is less than 1nm in diameter ($1\text{nm} = 10^{-9}\text{m}$).
- c) The particles of a solution can't be seen ever with a microscope.
- d) The particles of a solution pass through the filter paper. So, a solution can't be separated by filtration.
- e) The solutions are very stable. The particle of solute present in a solution do not separate out on keeping.
- f) A true solution does not scatter light i.e, it does not show Tyndall effect.

▪ **Types of Solution**

i) **Solution of Solid in a Solid**

- Metal alloys are the solutions of solids in solids.

For example:-

- ✓ Brass is a solution of zinc in copper.

ii) **Solution of Solid in a liquid**

- The solution which is made by solid in the liquid.

For example:-

- ✓ Salt and water solution, sugar and water solution.

iii) **Solution of liquid in a liquid**

- The solution which is made by liquid in liquid.

For example:-

- ✓ Vinegar is a solution of Acetic acid in water.

iv) **Solution of gas in liquid**

- The solution which is made by gas in the liquid.

For example:-

- ✓ Soda – water is a solution of CO_2 in water.

v) **Solution of gas in gas**

- The solution which is made by gas in gas.

For example:-

- ✓ Oxygen, argon, CO_2 , etc. are soluble in air.

• **Saturated Solution**

- A solution in which the maximum amount of solvent has been dissolved at a constant temperature, called saturated solution.

Note:

- If more solute is added, then its crystal will sit on the bottom of container.

For example:-

- ✓ 36g of salt (NaCl) in 100g of water at 20°C .
- ✓ 32g of Potassium Nitrate (KNO_3) in 100g of water at 20°C (293K).

• **Unsaturated Solution**

- That solution in which more solute can be dissolved at a given temperature is called unsaturated solution.

For example:-

- ✓ 20g of Salt (NaCl) in 100g of water.

• **Supersaturated solution**

- The amount of solute particles dissolved higher than required to prepare a saturated solution, called supersaturated solution.

For example:-

- ✓ 100g of water at 20°C and pressure of 1 A.T.M. can be dissolved a maximum amount of 34.7g of KCl; So, the solution is supersaturated solution.
✓ Honey.
✓ Sugar Syrup etc.

Special Note:

- A super saturated Solution is prepared by saturated solution with rising in temperature due to rising the temperature intermolecular space between particles increase. Thus, the particles of solute added trapes in it.
If temperature decreases and becomes at Normal temperature then the particles of super saturated solution separated in the form of crystal.

• **Solubility**

- The amount in grams of a solute dissolves in 100g of a solvent to make a saturated solution is called the solubility of the solute in the given solvent at a given temperature and pressure of the solution.

$$\therefore \text{Solubility} = \frac{\text{The amount of Solute (in g)}}{\text{The amount of Solvent (in g)}} \times 100$$

If the amount of Solute = m g

If the amount of Solvent = w g

$$\therefore \text{Solubility} = \frac{m}{w} \times 100$$

For example:-

- ✓ The solubility of KCl in water at 20°C and 1 A.T.M is 34.7g per 100g of water.

i.e, Solubility $= \frac{m}{w} \times 100$
 $= \frac{34.7}{100} \times 100 = 34.7\%$

Q. Find the solubility of 100g of NaCl dissolve in 1 Kg of water.

Ans Solubility $= \frac{m}{w} \times 100$
 $= \frac{100}{1000} \times 100 = 10\%$

$\therefore$ Solubility of NaCl = 10% OR 10g in 100g

▪ **Effect of temperature on the solubility of a solid (Solute)**

- In general cases, The solubility of a solid in a liquid increases with the rise in the temperature.

For example:-

- ✓ Potassium Nitrate and lead Nitrate are those solutes and their solution increases solubility when the temperature increases.

Note 1: Solubility depends on mainly two factors

Case1: The nature of the solute particles (Solid)

- The solubility increases in the case of solid potassium nitrate and lead Nitrate, but solubility decreases in the case of NaCl.

Case 2:

- Those substances which dissolved with water either endothermic or exothermic process.

For example:-

- In the case of lime (CaO).

Note 2: The solubility of a gas in a liquid decreases with rise in temperature but increases with increase in pressure.

- **Suspension**

- A suspension is a heterogeneous mixture in which small particles of a solid remains suspended throughout the mass of liquid or, gas.

For example:-

- ✓ Muddy water, Sand water, Chalk water, mixtures, paints etc.

- **Properties of a suspension**

- A suspension is a heterogeneous mixture.
- The particles in a suspension are large enough to be visible by un – necked eyes or, under an ordinary microscope.
- The particles in a suspension are too large remains in the suspension for a long period, further the intermolecular force between the particles are not so strong.
- The particles in a suspension can be separated from the liquid by filtration.
- The size of the particles is of the order of 10^{-5} cm or, larger.

- **Difference between the solution and Suspension**

TRUE Solution/Solution	Suspension
It is homogeneous.	It is heterogeneous.
It passes through filter paper.	The suspended particles don't pass through filter paper but liquid passes through filter paper.
The diameter of particles is of the order of 10^{-8} cm.	The diameter of the particle of the order of 10^{-5} cm.
Doesn't show Tyndall effect.	Shows Tyndall effect.
It does not show Brownian movement.	It shows Brownian motion.
It does not show electrophoresis.	It shows electrophoresis.
Particles can't be seen by naked eyes.	Particles can be seen by naked eyes.
It is transparent.	It is translucent.
Solute particles do not settle down on standing.	Solute particles settle down on standing.

- **Colloids or Colloidal Solution**

- A Colloids has properties that are intermediate between those of a true solution and a suspension.

- The colloids have particles that are larger than the particles of the true solution but smaller than those of a suspension.

For example:-

- ✓ Milk, ink, blood, toothpaste, starch solution, mist (Dispersion of water droplets in air), clouds, smoke, jellies etc.

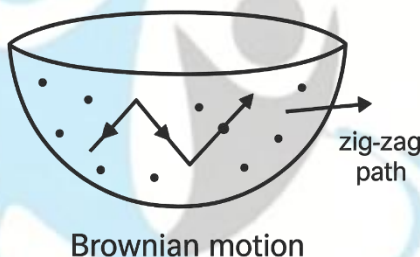
Note:

1. Colloids are either opaque material or, translucent material.
2. Toothpaste [Colgate, Pepsodent, Dant – Kanti, Close – up, Sensodyne, etc.], Cream [Fair lovely, Ponds – white beauty] are also gel but these are generally colloids.

▪ **Properties of Colloidal Solution**

- a) A Colloidal Solution is heterogeneous, the particles are visible under a powerful microscope.
- b) The particles in a colloids can pass through a filter paper.
- c) The particles in a colloids quite stable. They do not settle down at the bottom. When the colloids is allowed to stand still.
- d) The size of Colloidal particle ranging from 10^{-7} cm to 10^{-5} cm.
- e) Colloidal particles are in a state of Constant and rapid zig – zag motion called Brownian motion.

Brownian motion arises due to collision of colloidal particles with the particles of the solvent.

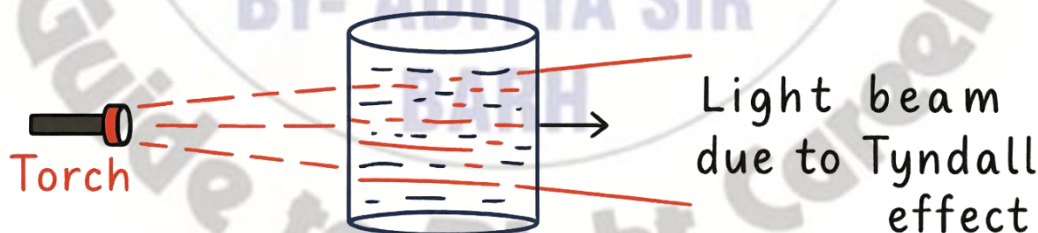


For example:-

- ✓ Pollen grain.

f) **Tyndall effect**

- The scattering of white light by colloidal particles is known as Tyndall effect.

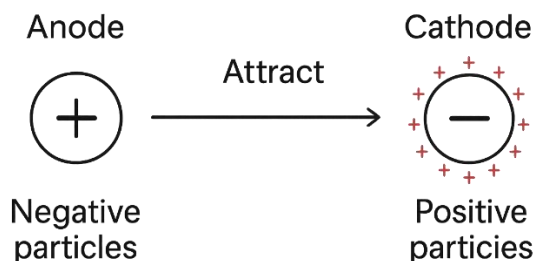


Note:

- True Solution do not show Tyndall effect.

g) **Electrophoresis**

- Colloidal particles carry charge.
- The movement of charged colloidal particles under the influence of an electric field, the charge colloidal particles move towards the oppositely charged electrodes and precipitated the phenomenon is also known as electrophoresis.



• **Differences between true solution & colloids**

TRUE Solution/Solution	Colloidal Solution
Its composition is homogeneous.	Its composition is heterogeneous.
It passes through the filter paper.	It can pass through the filter paper.
Diameter of the particles is of the order of 10^{-8} cm.	Particle's diameter lies between 10^{-7} cm to 10^{-5} cm.
It does not scatter light i.e, does not show Tyndall effect.	It shows Tyndall effect as Colloidal particles scatter light.
It does not show Brownian movement.	It shows the Brownian movement.
It does not show electrophoresis.	It shows maximum electrophoresis.
Particles cannot be seen.	Particles can be seen through a powerful microscope.
It is transparent.	It is translucent or opaque.
Solute particles do not settle down on keep stand still.	Colloidal particles do not settle down on keep stand still.

▪ **Types of Colloids**

1. **Sol**

- It is a mixture of solid and liquid.

For example:-

- ✓ Milk of magnesia $[\text{Mg}(\text{OH})_2]$, paint, ink etc.

2. **Emulsion**

- It is actually a mixture of liquid in liquid.

For example:-

- ✓ Hair cream, butter is dispersed in milk.

3. **Gel**

- It is a mixture of liquid in solid.

For example:-

- ✓ Cheese, Pepsodent, Colgate, Close – up etc.

4. **Aerosol**

- It is a mixture of solid in gas.

For example:-

- ✓ Smoke.

- It is also a mixture of liquid in gas.

For example:-

- ✓ Fog, Clouds, Perfume etc.

5. **Foam**

- It is the mixture in which a gas is dispersed in a liquid.

For example:-

- ✓ Shaving cream, bear foam, Soap bubble etc.

6. Solid Foam

- Solid Foam is that colloid in which a gas is dispersed in a solid medium.

For example:-

- ✓ Foam, rubber, Spong, bread etc.

▪ Types of Solution on the basis of amount of solute present in the solution

1. Concentrated Solution.
2. Dilute Solution.

1. Concentrated Solution

- The large amount of Solute dissolved in any solvent, called concentrated solution.
➤ The concentrated of solution is also called strength of the solution.

2. Dilute solution

- The small amount of solute dissolved in any solution is called dilute solution.
➤ The dilute solution is also called “low concentrated solution”.

Note:

- The concentration of solution a is expressed in the term of percentage (%).

Case 1:

- The concentration of solution by mass is defined as, “The mass of Solute in grams dissolved in 100g of Solution”.

$$\text{Concentration of Solution} = \frac{\text{Mass of Solute (g)}}{\text{Mass of Solution (g)}} \times 100$$

For example:-

- Q. Find the concentration of solution in which 10g of urea dissolved in 90g of water.

$$\begin{aligned}\text{Ans Concentration of Urea} &= \frac{10g}{(90 + 10)g} \times 100 \\ &= \frac{10g}{100g} \times 100 \\ \text{Concentration of Urea} &= 10\%\end{aligned}$$

Case 2:

- The concentration of solution by volume is defined as, of solute dissolved in 100 parts by volume of the solution.

$$\text{Concentration of Solution} = \frac{\text{Volume of Solute (in cm}^3\text{)}}{\text{Volume of Solution (in cm}^3\text{)}} \times 100$$

For example:-

- Q. Calculate the concentration of a solution in Volume % made when 56g of water is mixed with 0.17L of ethanol.

Ans ∴ Mass of water = 56g
& Density of water = 1g/CC

$$\therefore \text{Volume} = \frac{\text{Mass}}{\text{Density}}$$

$$\therefore \text{Volume} = \frac{56}{1.0} = 56 \text{ CC [cm}^3 = \text{CC]}$$

$$\begin{aligned}\therefore \text{Volume of ethanol} &= 0.17 \text{ L} \\ &= 0.17 \times 1000 \\ &= 170 \text{ CC}\end{aligned}$$

∴ Concentration of Solution in Volume

$$\begin{aligned}&= \frac{56}{170+56} \times 100 \\ &= \frac{56 \times 100}{226} = \frac{2800}{113} = 24.78\%\end{aligned}$$